[Generative Adversarial Networks] (GANs)

1000

↳ generate new images

(n ~~100~~ images) → (generate new images)

Complex Transformation

↓

Neural networks

Sample

$z \sim N(0, \sigma^2)$

given n number of mnist images, can I learn to generate more images from a simple normal dist.

NN Generator ⟷ NN Discriminator

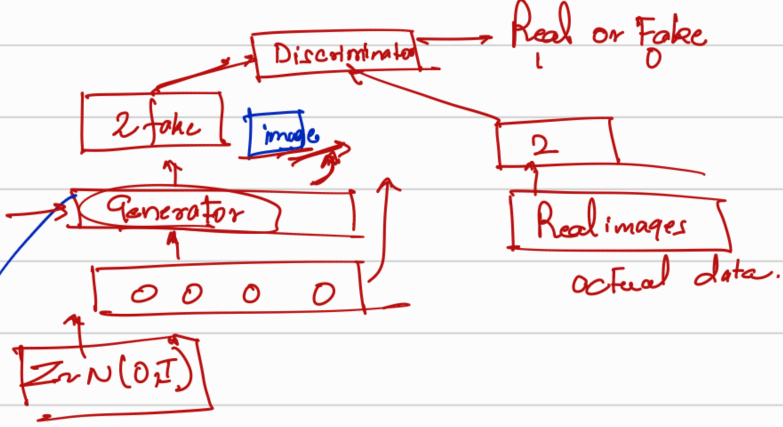
↳ Real Fake

generating the images from the normal dist

differentiating the image → real or fake.

① generate images such that discriminator finds it hard to discriminate

① to distinguish b/w fake and real in the best possible way.



Objective function of the generator

Neural network model

image → $G_\phi(z)$ → Parameters

NN

$D_\theta(G_\phi(z))$ → Parameters

Generator's objective ⇒ $\max \left[\log \left(D_\theta(G_\phi(z)) \right) \right]$

$$\max \int_z P(z) \times \log D_\theta G_\phi(z)$$

$$\min_\phi \int_z P(z) \times \left[1 - \log \left(D_\theta G_\phi(z) \right) \right]$$

$$\min \left[\log D_\theta G_\phi(z) \right]$$

$$\max \mathop{E}_{z \sim P(z)} \left[1 - \log D_\theta G_\phi(z) \right].$$

Discriminator's objective:

$$\max \mathop{E}_{x \sim \text{data}} \log \left(D_\theta(x) \right)$$

Real data

$$\max_\theta \left[\mathop{E}_{x \sim \text{data}} \log D_\theta(x) + \mathop{E}_{z \sim P(z)} \log \left(1 - D_\theta G_\phi(z) \right) \right]$$

$$\max \left[\log D_\theta G_\phi(z) \right] = \min \left[1 - \log D_\theta G_\phi(z) \right]$$

Combined objective function of the Generator & the discriminator ⇒

Gen Dis

$$\min_{\phi} \max_{\theta} \left[\mathbb{E}_{x\sim data} \log D_\theta(x) + \mathbb{E}_{z\sim p(z)} \log\left(1 - D_\theta(G_\phi(z))\right) \right]$$

x $\quad$ $\min \phi$

(how do we train such a model).

Step 1:

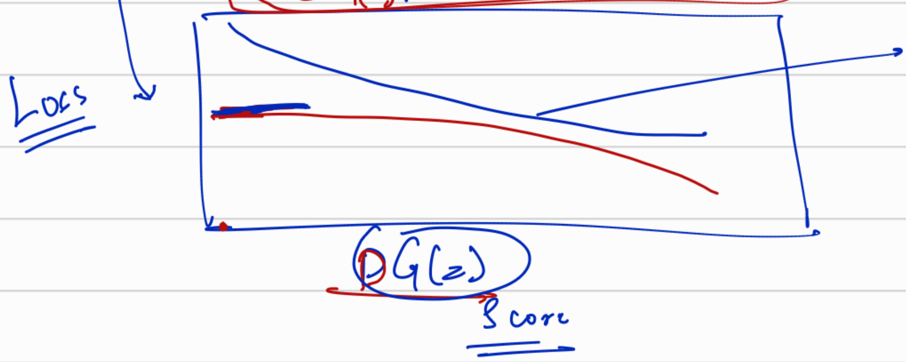
(i) Gradient ascent on the discriminator:

$$\rightarrow (i) \quad \max_{\theta} \left[\mathbb{E}_{x\sim pdata} \log D_\theta(x) + \mathbb{E}_{z\sim p(z)} \log\left(1 - D_\theta G_\phi(z)\right) \right]$$

- Gradient descent on the generator:

$$\min_{\phi} \mathbb{E}_{z\sim p(z)} \log\left(1 - D_\theta G_\phi(z)\right) \rightarrow \text{Loss}$$

$$\max_{\phi} \mathbb{E}_{z\sim p(z)} \log\left(D_\theta G_\phi(z)\right)$$



Procedure for GAN training.

for number of training iterations do:

for k steps:

- Sample minibatch of m noise samples $\{z^{(1)} \dots z^{(m)}\}$.
- Sample minibatch of m real examples $\{x^{(1)} \dots x^{(m)}\}$.
- update the discriminator by ascending the eq:

$$\nabla_\theta \frac{1}{m} \sum_{i=1}^{m} \Big[\underbrace{\log D_\theta(x^{(i)})}_{\text{Real}} + \underbrace{\log\big(1 - D_\theta G_\phi(z^{(i)})\big)}_{\text{Fake}} \Big]$$

end for

- Sample minibatch of m noise samples $\{z^{(1)} \dots z^{(m)}\}$.

update the generator by Stochastic gradient ascent:

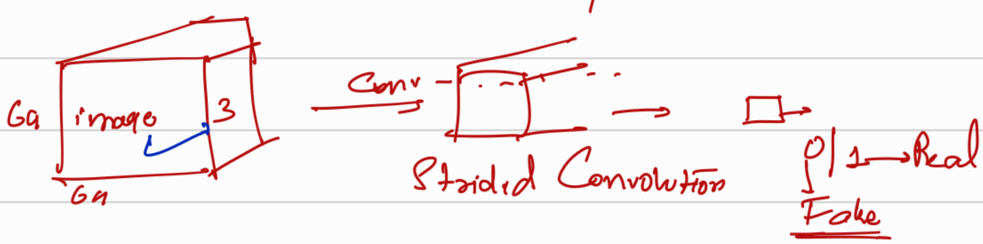
$$\nabla_\phi \frac{1}{m} \sum_{i=1}^{m} \Big[\log D_\theta G_\phi(z^{(i)}) \Big]$$

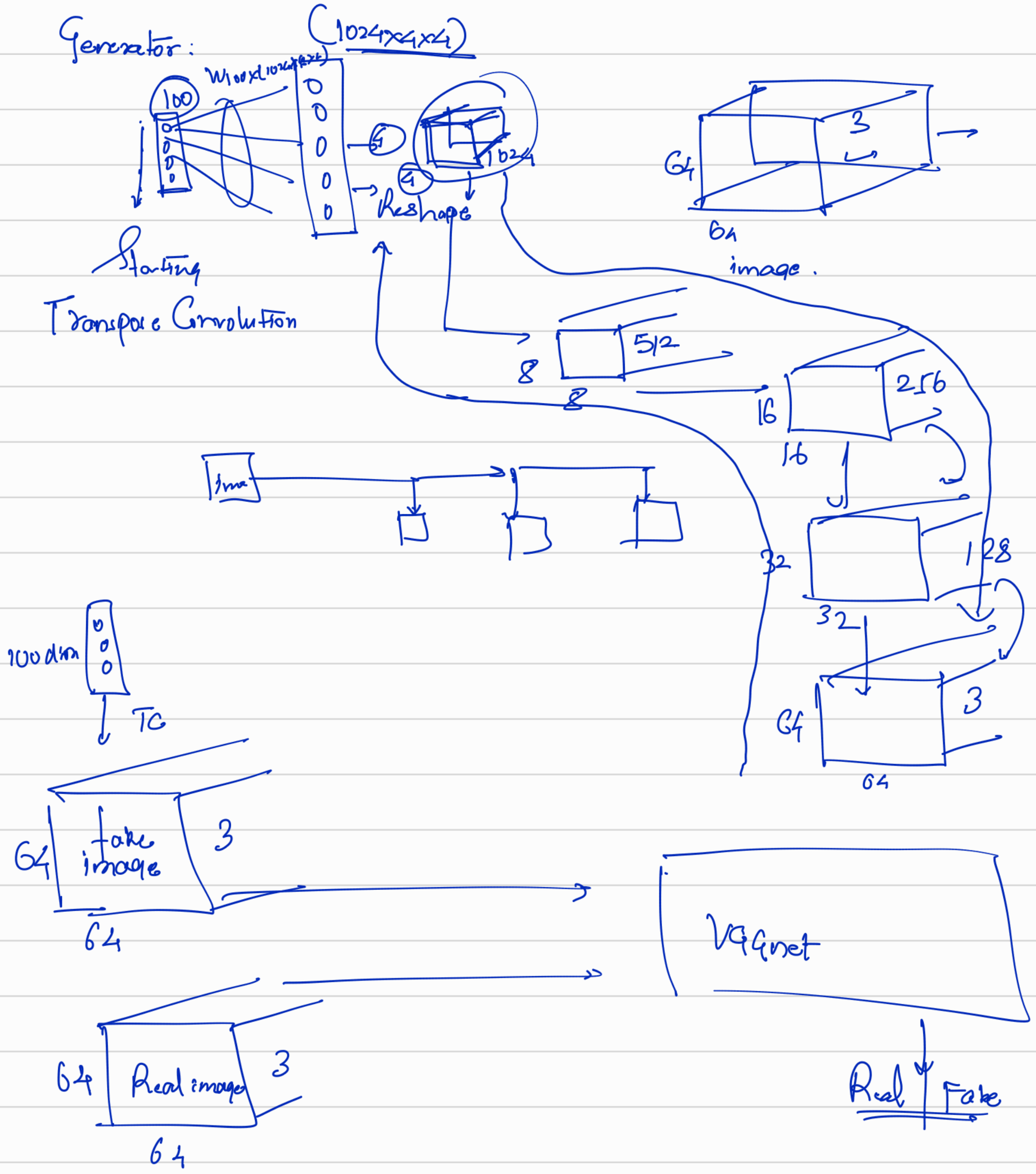
end for

DCGAN

Deep Convolution GANs

Discriminator architecture: (VGGnet / Resnet)

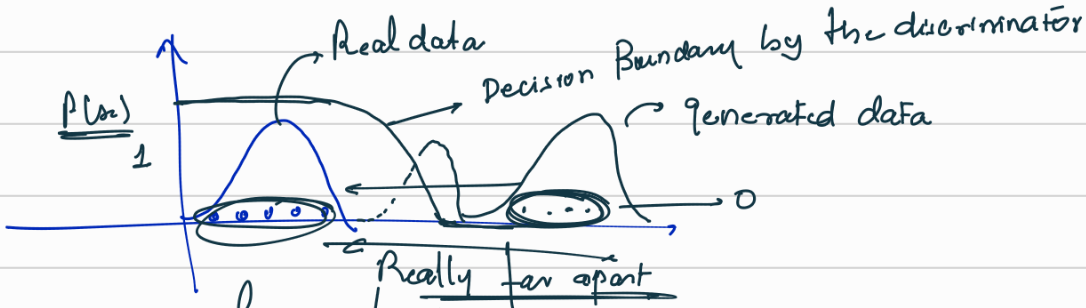




1. any Pooling layer has to be replaced with Strided Convolution
2. Use batchnorm for both the generator and the discriminator.
3. Remove any fully connected hidden layers →
4. Use Relu activation in generator for all layers and the output is tanh
5. Use Leaky Relu activation in the discriminator for all layers.

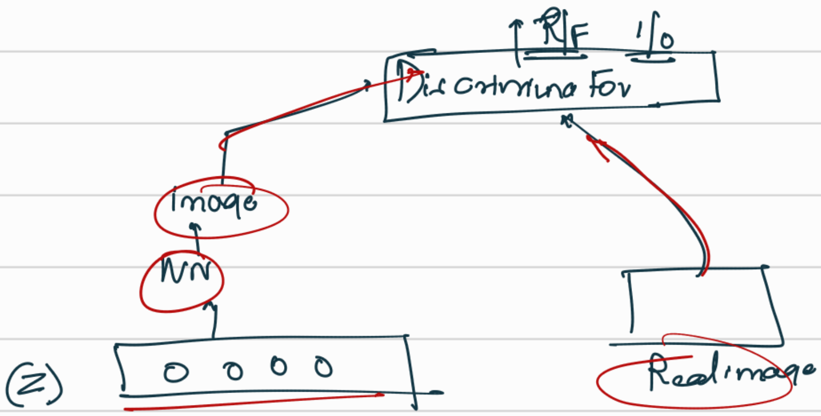
Improved GAN training

Wasserstein GAN

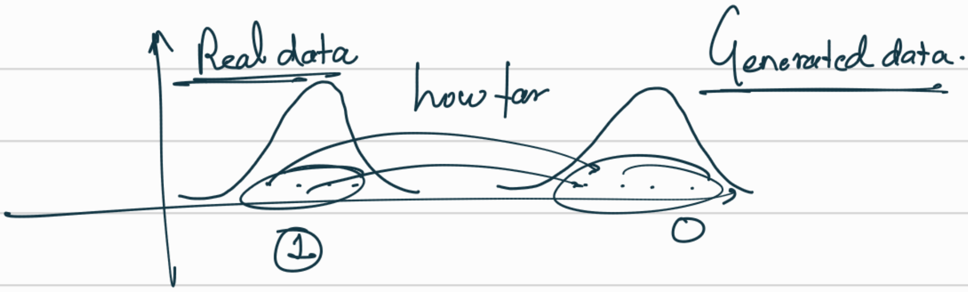


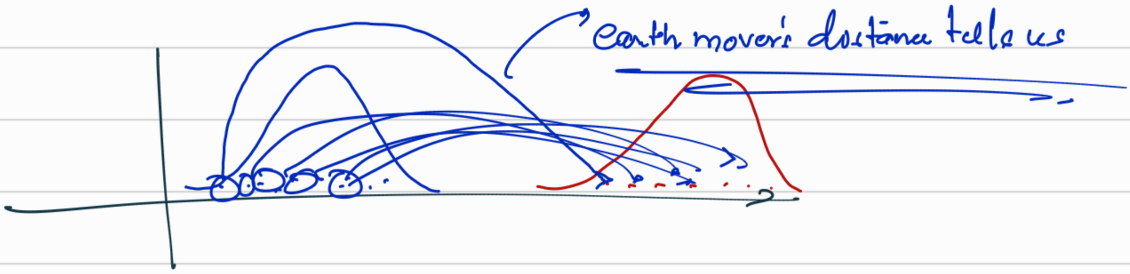
discriminator → learns faster.

WGANs



Earth mover's distance





Changes implemented in WGAN compared to normal GAN:

① Discriminator final layers ~~was~~ removed (sigmoid) and instead we used a linear layer.

② implementation of earth mover's distance

Cool theorem → Kantorovich - Rubinstein Duality.

Lipschitz constant

$$W(p_{data}, p_g) = E_{p_{data}} f[G(x)] - E_{p_g(x)}[f(x)]$$

$$\frac{1}{n} f(x) - \frac{1}{n} f(x)$$

real data $\quad p_g(x)$

Discriminator neural network

1 - Lipschitz scalar function → Discriminator Neural network

$$|f(x) - f(y)| \le |x - y|$$

If I change some data in my real image, the output does not change more than 1 times of the change in the image.

$g = 2x \rightarrow$ 1 unit

Lipschitz = 2

$y_1 = 2x_1$

$y_2 = 2x_2$

$$|y_2 - y_1| = 2|x_2 - x_1|$$

$$\frac{1}{n}\left[f_\theta(x_{real})\right] - \lambda\left(\|\nabla_x f_\theta(x)\|_2 - 1\right)^2 - \frac{1}{n} f_\theta[G(x)]$$

the change in the discriminator output value based on a unit change in the image norm

WGAN $\Rightarrow$

Style GANs

Code